

A

B

C

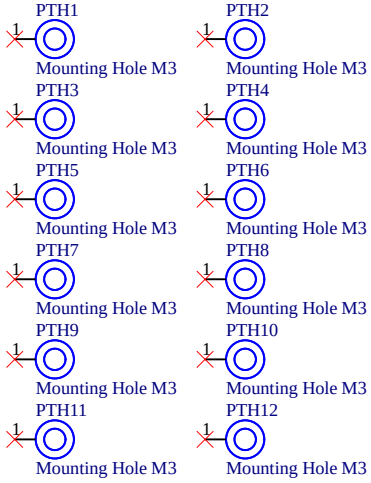
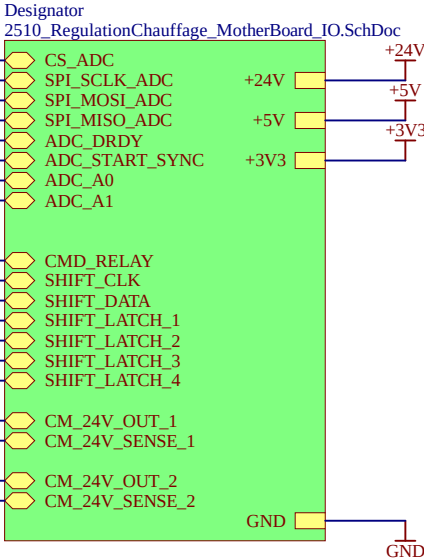
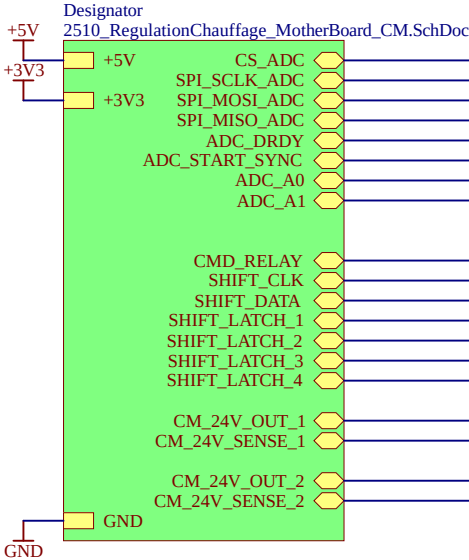
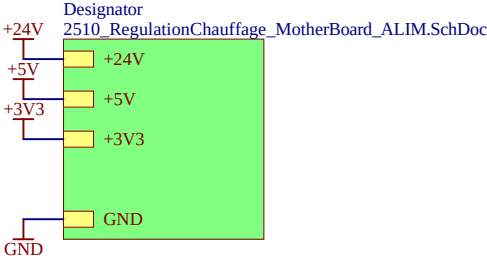
D

A

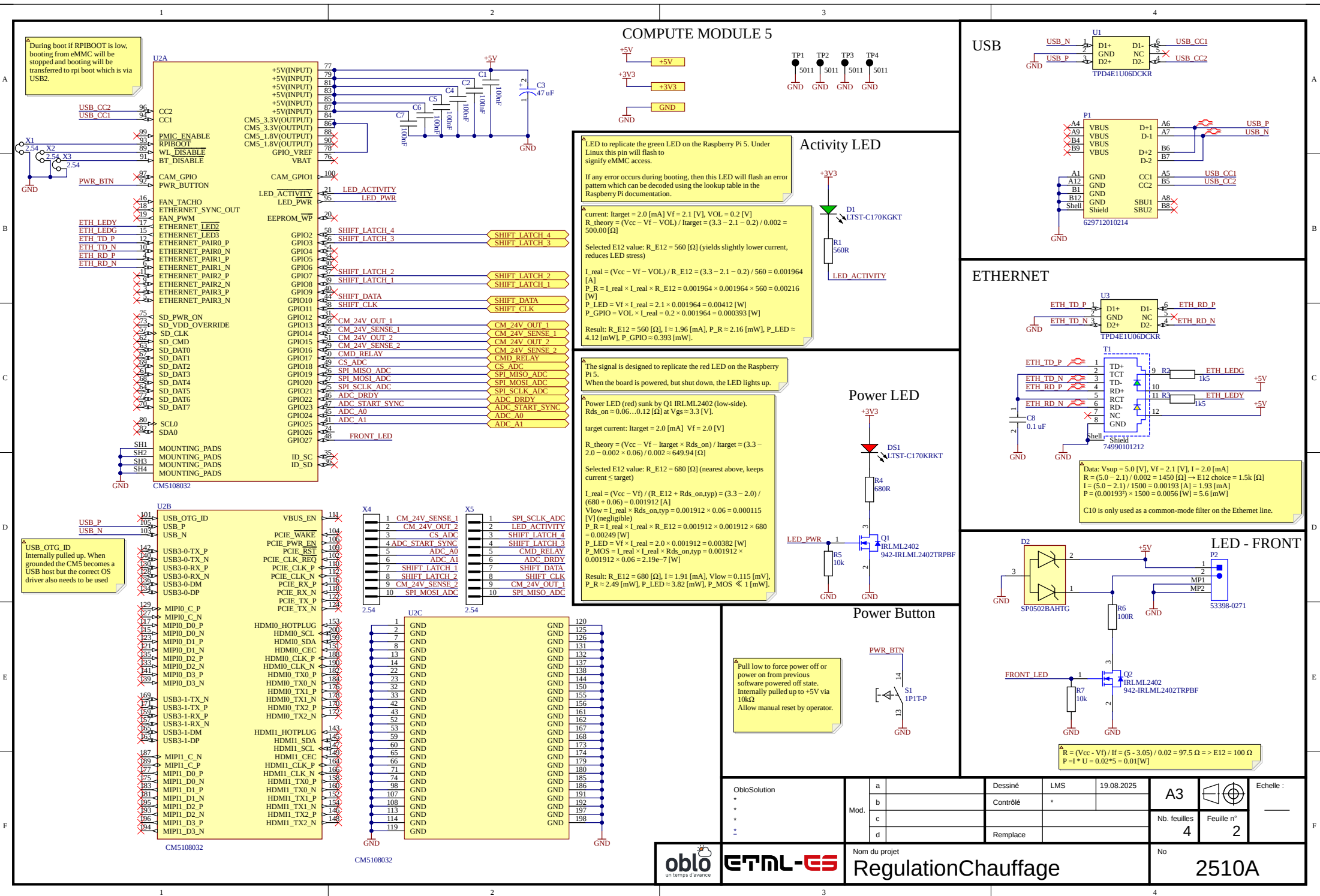
B

C

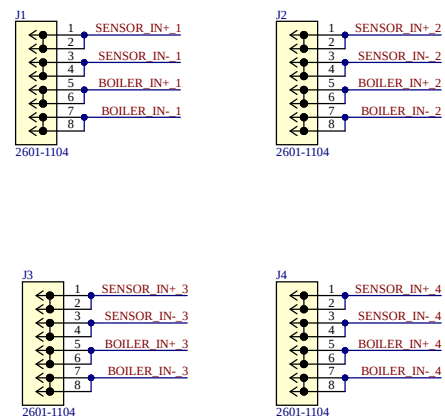
D



ObloSolution * * * * —	Mod.	a		Dessiné	LMS	19.08.2025	A4		Echelle : _____
		b		Contrôlé	*				
		c					Nb. feuilles	Feuille n°	
		d		Remplace			4	1	
	Nom du projet					 un temps d'avance	No		
	RegulationChauffage						2510A		

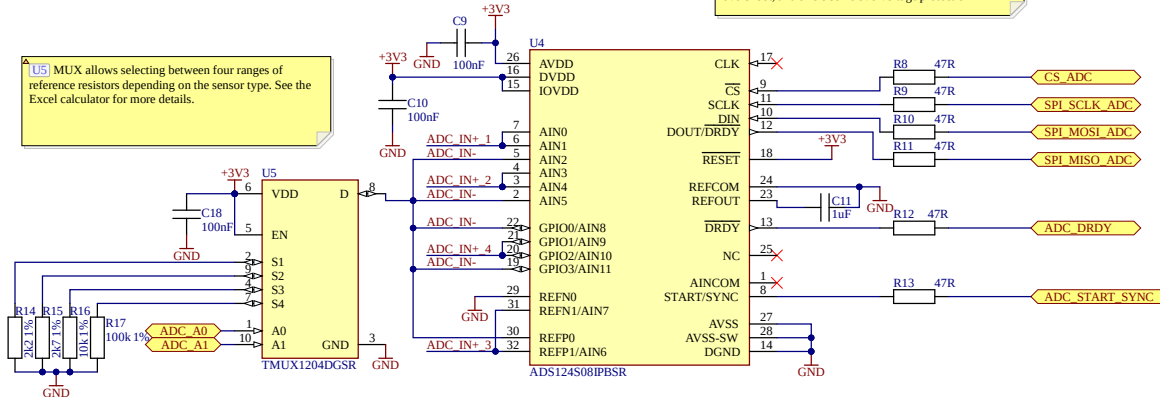


I/O Sensor / Boiler



ADC

U5 MUX allows selecting between four ranges of reference resistors depending on the sensor type. See the Excel calculator for more details.



R8 to R13 "47-Ω resistors in series with all digital input and output pins (CS, SCLK, DIN, DOUT/DRDY, and DRDY). This resistance smooths sharp transitions, suppresses overshoot, and offers some overvoltage protection"

U4 ADS124S08IPBSR – 4x 2-wire sensors

Reference: the internal 2.5 V reference is buffered and stabilized with 1 μF between REFOUT and REFCOM.

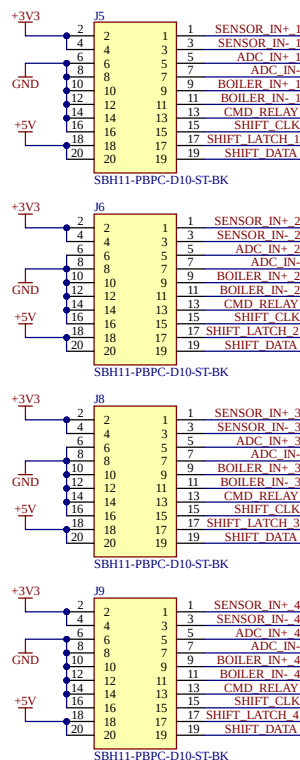
Digital interface: the ADC communicates with the Raspberry Pi CM5 over SPI (pins CS, SCLK, MOSI, MISO). Control signals: DRDY (data ready) and START/SYNC are routed for synchronization and conversion triggering. The internal clock is used, so the CLK pin is left unconnected.

I_{AVDD}(g=4): 140 μA typ; 155 μA max.
I_{DVDD}(conv): 225 μA typ; 300 μA max.
I_{REFInt}: 185 μA typ; 280 μA max.
I_{VBIAS} (sans charge): 10 μA.
I_{IDAC} overhead(250 μA): 20 μA typ; 35 μA max.

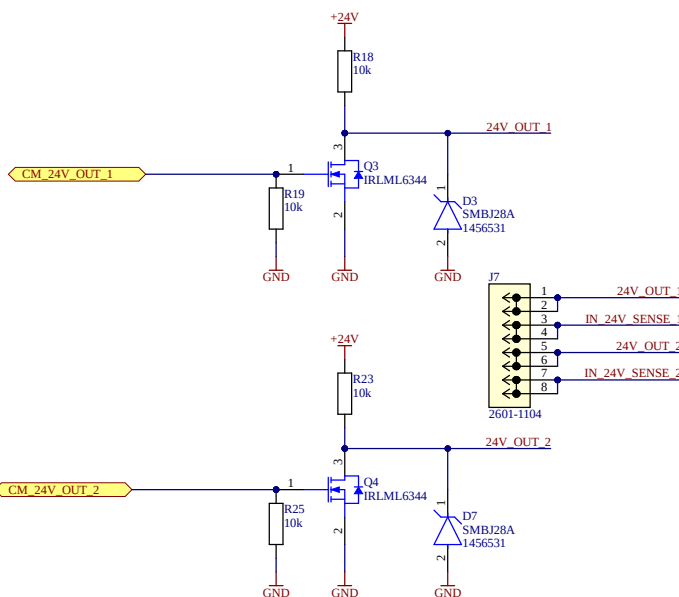
Données clé [μA] (datasheet):
I_{AVDD}(g=4,8) = 140 typ; 155 max
I_{DVDD}(conv, osc int) = 225 typ; 300 max
I_{IDAC} overhead(250 μA) = 20 typ; 35 max
I_{IDAC} = 250

Cas VBIAS non utilisé:
I_{total} typ = 140 + 225 + 20 + 250 = 635 [μA]
I_{total} max = 155 + 300 + 35 + 250 = 740 [μA]

I/O DAUGHTER BOARD



24V - 3V3 I/O



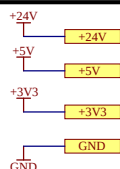
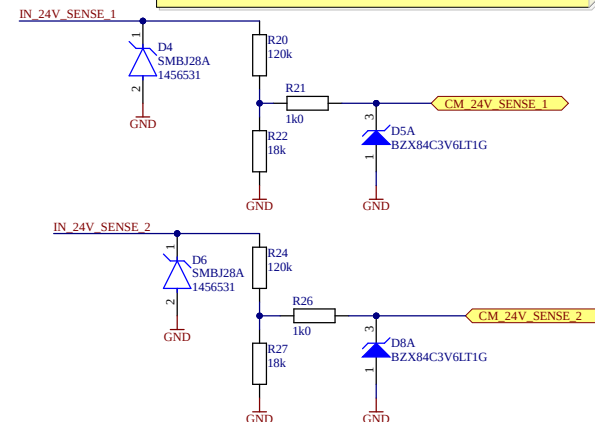
Single theoretical example, then multiplied. Two cases combined per channel:
- Output worst case (pull-up to 24 V held low through 10 kΩ)
- Input sense (24 V across divider 120 kΩ + 18 kΩ + 1 kΩ)

Output (per channel):
 $I_{out} = 24 / 10000 = 0.0024 \text{ A}$
 $P_{out} = I_{out} * I_{out} * 10000 = 0.0024 * 0.0024 * 10000 = 0.0576 \text{ W}$

Input (per channel):
 $I_{in} = 24 / (120000 + 18000 + 1000) = 24 / 139000 = 0.000173 \text{ A}$
 $P_{in} = 24 * I_{in} = 24 * 0.000173 = 0.00415 \text{ W}$

Combined per channel (output + input):
 $I_{ch} = 0.0024 + 0.000173 = 0.002573 \text{ A}$
 $P_{ch} = 0.0576 + 0.00415 = 0.06175 \text{ W}$

Totals for 2 identical channels:
 $I_{total} = 2 * 0.002573 = 0.005146 \text{ A}$
 $P_{total} = 2 * 0.06175 = 0.1235 \text{ W}$



ObloSolution
*
*
*

Mod.
a
b
c
d

Dessiné
Contrôlé
Remplace

LMS
*
19.08.2025

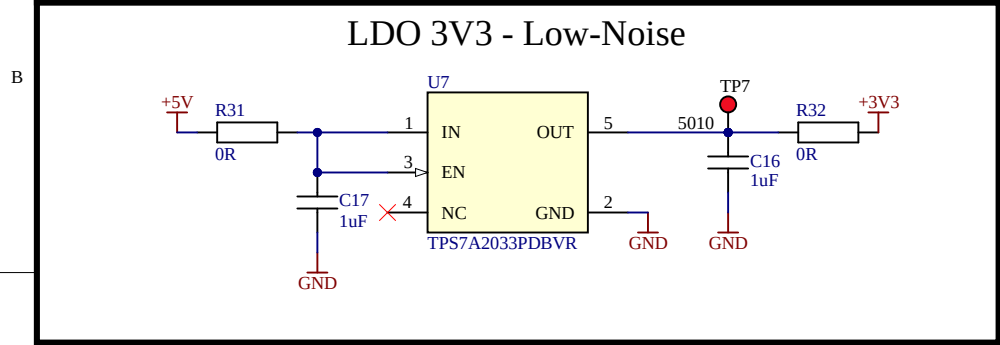
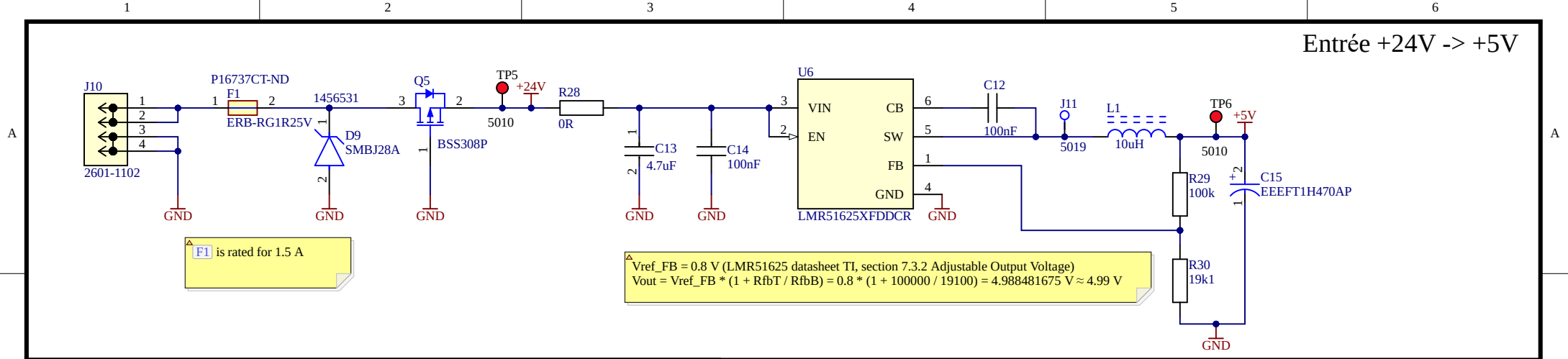
A3
Nb. feuilles
4
Feuille n°
3
Echelle :
—

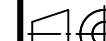
oblo
un temps d'avance

ETML-ES

Nom du projet
Regulation Chauffage

No
2510A



ObloSolution * * * * —	Mod.	a		Dessiné	LMS	19.08.2025	A4		Echelle : _____
		b		Contrôlé	*				
		c					Nb. feuilles	Feuille n°	
		d		Remplace			4	4	
	Nom du projet					 un temps d'avance	No		
	RegulationChauffage						2510A		

Line #	Name	Description	Designator	Revision ID	Revision Status	Quantity	TargetPrice	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	5019		J11			1	0	Keystone	5019	Volume Production	DigiKey	36-5019CT-ND	0.37	0.37
	CM5108032		U2			1	0	Amphenol Communications Solutions	10164227-1004A1RLF	Volume Production	DigiKey	609-10164227-1004A1RLFCT-ND	2.55	2.55
	IRLML2402		Q1, Q2			2	0	Infineon	IRLML2402TRPBF	Obsolete	Mouser	942-IRLML2402TRPBF	0.333	0.666
	Mounting Hole M3		PTH1, PTH2, PTH3, PTH4, PTH5, PTH6, PTH7, PTH8, PTH9, PTH10, PTH11, PTH12			12	0							
	second connector CM5		U2-21			1	0	Amphenol Communications Solutions	10164227-1004A1RLF	Volume Production	DigiKey	609-10164227-1004A1RLFCT-ND	2.55	2.55
	TPS7A2033PDBVR	300mA ultra-low-noise low-IQ low-dropout (LDO) linear regulator with high PSRR 5-SOT-23 - 40 to 125	U7			1	0	Texas Instruments	TPS7A2033PDBVR	Volume Production	Mouser	595-TPS7A2033PDBVR	0.31	0.31
	74990101212	74990101212 Telecom and Datacom Connector, 8 Contact(s), Female, Right Angle, Solder Terminal, Locking	T1			1	0	Würth Elektronik	74990101212		Farnell	2749150	6.7	6.7
	LMR51625XFDDCR	Abaisseur de tension Régulateur à découpage IC Positif Réglable 0,8V 1 Sortie 2,5A SOT-23-6 mince, TSOT-23-6	U6			1	0	Texas Instruments	LMR51625XFDDCR	Volume Production	DigiKey	296-LMR51625XFDDCRCT-ND	2.03	2.03
	BSS308P	Canal P 30 V 2 A (Ta) 500mW (Ta) Montage en surface PG-SOT23	Q5			1	0	Infineon	BSS308PEH6327XTSA1	Volume Production	Farnell	2432719	0.48594	2.43
	47 uF	CAP ALUM 47UF 20% 50V SMD	C3			1	0	Panasonic	EEFT1J470AP	Volume Production	Farnell	1868425	0.64428	0.64428
	EEFT1H470AP	CAP ALUM 47UF 20% 50V SMD	C15			1	0	Panasonic	EEFT1H470AP	Volume Production	Farnell	1868425	0.64428	0.64428
	0.1uF	CAP CER 0.1uF 10V X5R 0805	C8			1	0	Yageo Group	CC0805KRX7R9BB104	Volume Production	Farnell	3019949	0.03003	0.30028
	4.7uF	CAP CER 4.7UF 10V X5R 0805	C13			1	0	Yageo Group	CC0805KKX5R6BB475	Volume Production	DigiKey	311-1863-1-ND	0.2	0.2
	1uF	CAPC 1uF X5R 25V 10% 2012 (0805 INCH)	C11, C16, C17			3	0	TDK	C2012X5R1E105K125AA	Volume Production	DigiKey	445-1419-1-ND	0.1866	0.55981
	100nF	CAPC 100nF X7R 50V 10% 2012 (0805 INCH)	C1, C2, C4, C5, C6, C7, C9, C10, C12, C14, C18			11	0	KYOCERA AVX	08055C104KAT2A	Volume Production	Farnell	499687	0.03838	0.42214
	SBH11-PBPC-D10-ST-BK	Connector Header Through Hole 20 position 0.100 (2.54mm)	J5, J6, J8, J9			4	0	Sullins	SBH11-PBPC-D10-ST-BK	Volume Production	DigiKey	S9172-ND	0.41	1.64
	2601-1102	Description	J10			1	0	WAGO	2601-1102		Farnell	4126046	1.91	1.91
	2601-1104	Description	J1, J2, J3, J4, J7			5	0	WAGO	2601-1104		Farnell	4126048	3.54	17.69
	ADS124S08IPBSR	IC ADC 24BIT SIGMA-DELTA 32LQFP	U4			1	0	Texas Instruments	ADS124S08IPBSR	Volume Production	DigiKey	296-45170-1-ND	11.78	11.78
	2.54	JUMPER 2.54 1x02	X1, X2, X3			3	0	Würth Elektronik	61303611121		Mouser	710-61303611121	1.38	4.14
	LTST-C170KRKT	LED Red, 75 mW, 54 mcd, -55 to 85 degC, 2-Pin SMD, RoHS, Tape and Reel	DS1			1	0	Lite-On	LTST-C170KRKT	Volume Production	DigiKey	160-1415-1-ND	0.16	0.16
	53398-0271	Male Header, Pitch 1.25 mm, 1 x 2 Position, Height 4.7 mm, -40 to 85 degC, RoHS, Tape and Reel	P2			1	0	Molex	53398-0271	Volume Production	DigiKey	WM7606CT-ND	0.54	0.54
	ERB-RG1R25V	Micro Chip Fuse, Size [LÄ—W] (EIA) (mm): 3.2Ä—1.6 (EIA-1206), Rated Current (A): 1.25, Internal R [at 25Ä,f max.] (mÄ, I): 175, Rated Voltage (Open Circuit Voltage) [DC] (V): 63, Interrupting Rating (A): 50, Category Temp. Range (Ä°C): -40 to +125	F1			1	0				Digikey	P16737CT-ND		
	IRLML6344	MOSFET N-CH 30V 5A MICRO3/SOT23	Q3, Q4			2	0	Infineon	IRLML6344TRPBF	Volume Production	Farnell	1857299	0.39373	1.97
	TMUX1204DGSR	MULTIPLEXER	U5			1	0	Texas Instruments	TMUX1204DGSR	Volume Production	Mouser	595-TMUX1204DGSR	0.57	0.57
	Raspberry Pi SC1570	Ordinateur monocarte 2.4GHz 4 Cœur 8Go/0Go RAM ARM® Cortex®-A76	U2(Externe)1			1	0	Raspberry Pi	CM5008016		Newark	20AM3761	80	80
	2.54	PIN HEADER 2.54 1x10	X4, X5			2	0	Würth Elektronik	61300411121		Mouser	710-61300411121	0.17	0.34
	TPD4E1U06DCKR	Quad 0.8-pF, 5.5-V, Ä±15-kV ESD protection diode in SOT-23 & SC70 packages 6-SC70 -40 to 125	U1, U3			2	0	Texas Instruments	TPD4E1U06DCKR	Volume Production	DigiKey	296-39347-6-ND	0.37	0.74
	0R	RES 0ohm ±1% 0.125W 2012 (0805 INCH)	R28, R31, R32			3	0	Vishay	CRCW0805000020EA	Volume Production	Newark	52K9721	0.022	0.066
	1k0	RES 1.0kohm ±1% 0.125W 2012 (0805 INCH)	R21, R26			2	0	Yageo Group	RC0805FR-071KL	Volume Production	Newark	82K7536	0.004	0.008
	1k5	RES 1.5kohm ±1% 0.125W 2012 (0805 INCH)	R2, R3			2	0	Yageo Group	RC0805FR-071K5L	Volume Production	DigiKey	311-1.50KCRCT-ND	0.01633	0.16328
	2k2 1%	RES 2.2K OHM 1% 1/8W 0805 RC0805FR-072K2L	R14			1	0	Yageo Group	RC0805FR-072K2L	Volume Production	Farnell	9237534	0.01221	0.12211
	2k7 1%	RES 2.7K OHM 1% 1/8W 0805 RC0805FR-102K7L	R15			1	0	Yageo Group	RC0805FR-102K7L	Volume Production	DigiKey	13-RC0805FR-102K7LCT-ND	0.1	0.1
	10k	RES 10kohm ±1% 0.125W 2012 (0805 INCH)	R5, R7, R18, R19, R23, R25			6	0	Yageo Group	AC0805FR-0710KL	Volume Production	Newark	49AK0724	0.005	0.03
	18k	RES 18kohm ±1% 0.125W 2012 (0805 INCH)	R22, R27			2	0	Yageo Group	RC0805FR-0718KL	Volume Production	Newark	49AK3089	0.006	0.012
	19k1	RES 19k1 OHM 5% 1/4W 0805	R30			1	0	Yageo Group	RC0805FR-0719K1L	Volume Production	Farnell	4061021	0.10965	0.10965
	47R	RES 47ohm ±1% 0.125W 2012 (0805 INCH)	R8, R9, R10, R11, R12, R13			6	0	Yageo Group	RC0805FR-0747RL	Volume Production	Newark	98K7855	0.004	0.024
	100k 1%	RES 100K OHM 1% 1/8W 0805 AC0805FR-13100KL	R17			1	0	Yageo Group	AC0805FR-13100KL	Volume Production	Farnell	4013022	0.02006	0.2006
	100k	RES 100kohm ±1% 0.125W 2012 (0805 INCH)	R29			1	0	Yageo Group	RC0805FR-07100KL	Volume Production	Mouser	603-RC0805FR-07100KL	0.12	0.12
	100R	RES 100ohm ±1% 0.125W 2012 (0805 INCH)	R6			1	0	Yageo Group	RC0805FR-07100RL	Volume Production	RS	6185135	0.00454	0.00454
	120k	RES 120kohm ±1% 0.125W 2012 (0805 INCH)	R20, R24			2	0	Yageo Group	RC0805FR-07120KL	Volume Production	Newark	49AK3045	0.004	0.008
	560R	RES 560ohm ±1% 0.125W 2012 (0805 INCH)	R1			1	0				ETML	FAB.0805 R 560R		
	680R	RES 680ohm ±1% 0.125W 2012 (0805 INCH)	R4			1	0				ETML	FAB.0805 R 680R		
	10k 1%	RES SMD 10K OHM 0.1% 1/8W 0805 AC0805FR-0710KL	R16			1	0	Yageo Group	AC0805FR-0710KL	Volume Production	Farnell	3495316	0.01782	0.17818
	10uH	Semi-Shielded Power Inductor, 10 uH, +/- 20%, 2.1 A, -40 to 125 degC, 2-Pin SMD, RoHS, Tape and Reel SRN3015-100M	L1			1	0	Bourns	SRN3015-100M	Volume Production	Farnell	2061634	0.40993	2.06
	1P1T-P	SW 600X600X700 24V 50mA THT	S1			1	0	TE Connectivity	1-1825910-4	Volume Production	Newark	95M4263	0.019	0.019
	5011	Test Point, Black, Through Hole, RoHS, Bulk	TP1, TP2, TP3, TP4			4	0	Keystone	5011	Volume Production	Newark	88H2535	0.137	0.548
	5010	Test Point, Red, Through Hole, RoHS, Bulk	TP5, TP6, TP7			3	0	Keystone	5010	Volume Production	Mouser	534-5010	0.37	1.11
	SP0502BAHTG	TVS DIODE 5.5V 8.5V SOT23-3	D2			1	0	Littelfuse	SP0502BAHTG	Volume Production	Farnell	1827631	0.69775	0.69775
	SMBJ28A	TVS DIODE 28V 45.4V DO214AA	D3, D4, D6, D7, D9			5	0	Littelfuse	SMBJ28A	Volume Production	Farnell	1456531	0.25169	1.26
	LTST-C170KGKT	Ultra Bright AlInGaP Chip LED, 75 mW, -55 to 85 degC, 2-Pin SMD (0805), RoHS, Tape and Reel	D1			1	0	Lite-On	LTST-C170KGKT	Volume Production	DigiKey	160-1414-1-ND	0.18	0.18
	629712010214	USB 2.0 Type C Plug Vertical SMT	P1			1	0	Würth Elektronik	629712010214		DigiKey	732-629712010214CT-ND	4.55	4.55
	BZX84C3V6LT1G	Zener Voltage Regulator, 225 mW, 3-Pin SOT-23, Pb-Free, Tape and Reel	D5, D8			2	0	onsemi	BZX84C3V6LT1G	Volume Production	Farnell	2317468	0.11339	0.56694